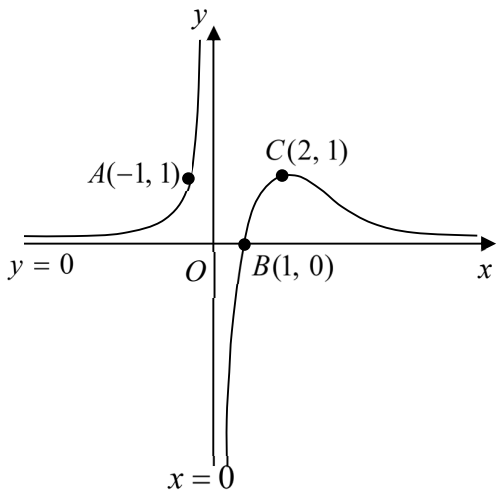
RAFFLES INSTITUTION
H2 Mathematics (9758)
2025 Year 5

Additional Practice Questions for Chapter 8B: Curve Transformation

Transformations

1

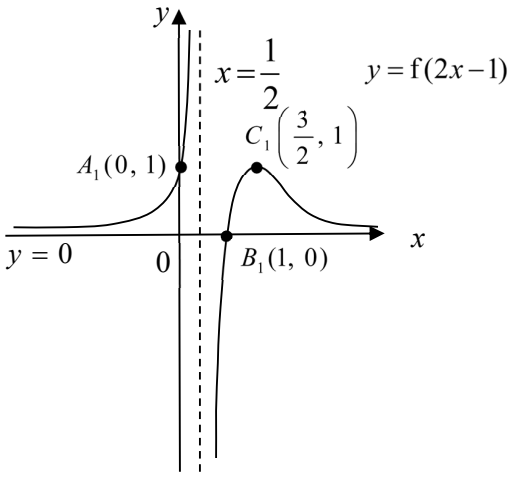


The diagram shows the graph of $y = \mathrm{f}(x)$. The curve passes through the points $A(-1,1)$ and $B(1,0)$, and has a maximum point at $C(2,1)$. The curve has asymptotes $x=0$ and $y=0$. Sketch, on separate diagrams, the graphs of

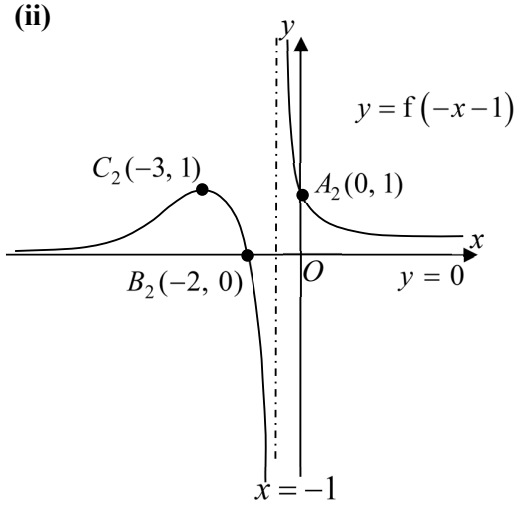
(i) $y = \mathrm{f}(2x-1)$, **(ii)** $y = \mathrm{f}(-x-1)$, **(iii)** $y = 2\mathrm{f}(2x)+1$.

In each case, state the coordinates of the points corresponding to *A*, *B*, and *C*, and the equations of the asymptotes, whenever applicable. **[9]**

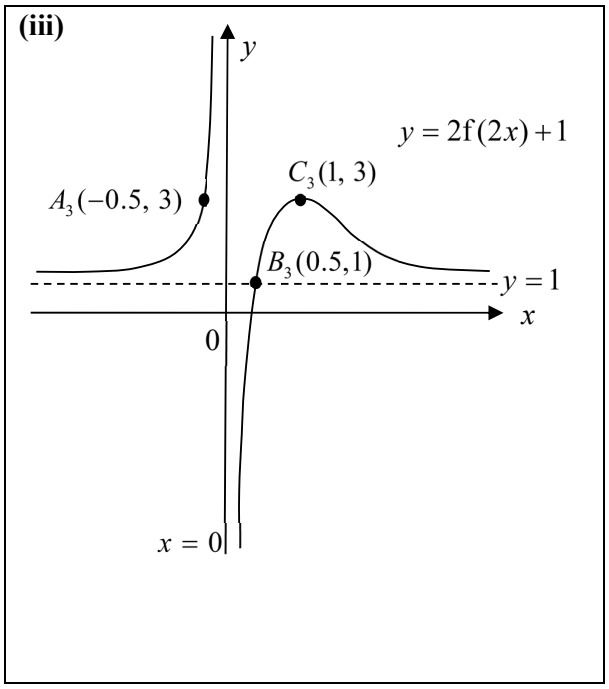
(i)



(ii)



(iii)



2 State precisely how each of the following curves may be obtained by applying a sequence of geometrical transformations from the given starting curve.

(a) $y = 3\mathrm{e}^{x+2}$ (starting curve : $y = \mathrm{e}^{x}$),

(b) $y = 1 - 2\sqrt{1-x}$ (starting curve : $y = \sqrt{x}$). **[7]**

(a)

$y = \mathrm{e}^{x}$ $\xrightarrow{\text{Replace } x \text{ with } x+2}$ $y = \mathrm{e}^{x+2}$ $\xrightarrow{\text{Replace } y \text{ with } \frac{y}{3}}$ $y = 3\mathrm{e}^{x+2}$

Translate the graph of $y = \mathrm{e}^{x}$ by 2 units in the negative x-direction to obtain the graph of $y = \mathrm{e}^{x+2}$.	Scale the graph of $y = \mathrm{e}^{x+2}$ parallel to the y-axis by a factor of 3 to obtain the graph of $y = 3\mathrm{e}^{x+2}$.

(b)

$y=\sqrt{x}$ $\xrightarrow{\text{Replace } x \text{ with } x+1}$ $y=\sqrt{1+x}$ $\xrightarrow{\text{Replace } x \text{ with } -x}$ $y=\sqrt{1-x}$

Translate the graph of $y=\sqrt{x}$ by 1 unit in the negative x-direction to obtain the graph of $y=\sqrt{1+x}$.

Reflect the graph of $y=\sqrt{1+x}$ in the y-axis to obtain the graph of $y=\sqrt{1-x}$.

$\xrightarrow{\text{Replace } y \text{ with } -y}$ $y=-\sqrt{1-x}$ $\xrightarrow{\text{Replace } y \text{ with } \frac{y}{2}}$ $y=-2\sqrt{1-x}$

Reflect the graph of $y=\sqrt{1-x}$ in the x-axis to obtain the graph of $y=-\sqrt{1-x}$.

Scale the graph of $y=-\sqrt{1-x}$ parallel to the y-axis by a factor of 2 to obtain the graph of $y=-2\sqrt{1-x}$

$\xrightarrow{\text{Replace } y \text{ with } y-1}$ $y=1-2\sqrt{1-x}$

Translate the graph of $y=-2\sqrt{1-x}$ by 1 unit in the positive y-direction.

$y=-\sqrt{1-x}$

Alternative for the first 2 transformations for (b):

$y=\sqrt{x} \to y=\sqrt{-x} \to y=\sqrt{-(x-1)}=\sqrt{1-x}$

Description:

Reflect the graph of $y=\sqrt{x}$ in the y-axis to obtain the graph of $y=\sqrt{-x}$, followed by translation of 1 unit in the positive x-direction to obtain the graph of $y=\sqrt{-(x-1)}=\sqrt{1-x}$.

Alternative for the last 3 transformations for (b):

Translate the graph of $y=\sqrt{1-x}$ by $\frac{1}{2}$ units in the negative y direction to obtain the graph of $y=-\frac{1}{2}+\sqrt{1-x}$, followed by reflection in the x-axis to obtain $y=\frac{1}{2}-\sqrt{1-x}$ and a scaling by factor 2 parallel to the y-axis to obtain $y=1-2\sqrt{1-x}$

3 The transformations A, B and C are as follows:

A: A translation of 1 unit in the negative y-direction;

B: A stretching parallel to the y-axis (with x-axis invariant) by a factor of 2;

C: A reflection about the x-axis.

A curve undergoes in succession, the transformations A, B and C, and the equation of the resulting curve is $y = -2x^2 - 6$.

(i) Determine the equation of the curve before the transformations were effected. **[3]**

Suppose the curve undergoes in succession, the above transformations in the reverse order (i.e. C, B and A).

(ii) Determine the equation of the resulting curve. **[3]**

(iii) Suggest one amendment to the above transformations for the reverse order such that the resulting equation is still $y = -2x^2 - 6$. **[1]**

(i)

$$y = -2x^2 - 6 \xrightarrow{\text{just before } C} -y = -2x^2 - 6 \xrightarrow{\text{just before } B} \frac{y}{1/2} = 2x^2 + 6$$

$$y = -2x^2 - 6 \xrightarrow{\text{just before } C} y = 2x^2 + 6 \xrightarrow{\text{just before } B} y = \tfrac{1}{2}\left(2x^2 + 6\right) = x^2 + 3$$

$$\xrightarrow{\text{just before } A} y - 1 = x^2 + 3$$

$$y = x^2 + 4$$

Thus, the equation of the curve before the transformations were effected is $y = x^2 + 4$.

Alternatively,

$$y = \mathrm{f}(x) \xrightarrow{A} y + 1 = \mathrm{f}(x) \xrightarrow{B} \frac{y}{2} + 1 = \mathrm{f}(x) \xrightarrow{C} \frac{-y}{2} + 1 = \mathrm{f}(x), \text{ i.e. } y = 2[1 - \mathrm{f}(x)]$$

So $-2x^2 - 6 = 2[1 - \mathrm{f}(x)] \Rightarrow 1 - \mathrm{f}(x) = -x^2 - 3 \Rightarrow \mathrm{f}(x) = x^2 + 4$

Thus, the equation of the curve before the transformations were effected is $y = x^2 + 4$.

(ii)

$$y = x^2 + 4 \xrightarrow{C} -y = x^2 + 4 \xrightarrow{B} -\frac{y}{2} = x^2 + 4 \xrightarrow{A} -\frac{y+1}{2} = x^2 + 4, \text{ i.e. } y = -2x^2 - 9$$

The equation of the resulting curve is $y = -2x^2 - 9$.

(iii) Amend A to a translation of 2 units in the positive y-direction.

4 9758/2021/01/Q6

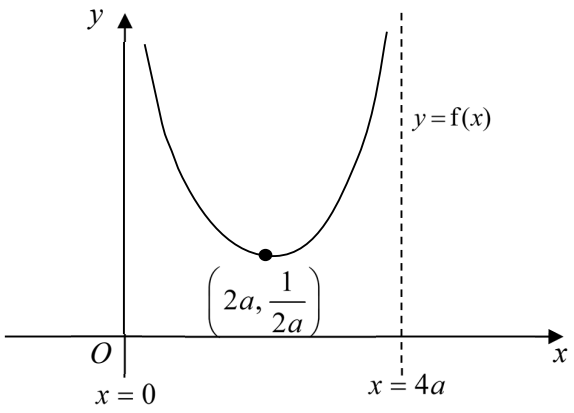
A curve C has equation $y = \dfrac{1}{\sqrt{4ax - x^2}}$, where $a > 0$.

(a) Sketch C and give the equations of any asymptotes, in terms of a where appropriate. **[4]**

(b) Find the smallest possible value of y in terms of a. **[1]**

(c) Describe the transformation that maps the graph of C onto the graph of $y = \dfrac{1}{\sqrt{4a^2 - x^2}}$. **[3]**

(a) $y = \dfrac{1}{\sqrt{4ax - x^2}} = \dfrac{1}{\sqrt{x(4a - x)}}$,



(b) When $x = 2a$, $y = \dfrac{1}{2a}$.

The smallest possible value of y is $y = \dfrac{1}{2a}$.

(c) $y = \dfrac{1}{\sqrt{4ax - x^2}}$

$= \dfrac{1}{\sqrt{-(x^2 - 4ax)}}$

$= \dfrac{1}{\sqrt{-\left[(x-2a)^2 - (2a)^2\right]}}$

$= \dfrac{1}{\sqrt{4a^2 - (x-2a)^2}}$

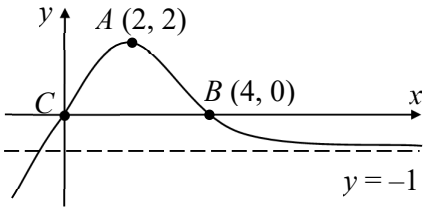
$\downarrow$ Replace x by $(x + 2a)$

$y = \dfrac{1}{\sqrt{4a^2 - x^2}}$

Translate the graph of C $2a$ units in the negative x-direction.

Transformations involving Modulus Functions

5 The graph of $y = \mathrm{f}(2x) - 1$ is shown.



Sketch on separate diagrams, the graphs of

(i) $y = -2\,\mathrm{f}(2x)$,

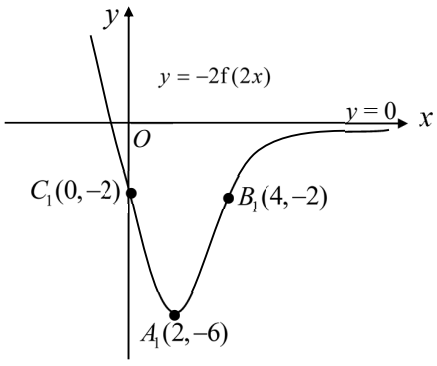
(ii) $y = \mathrm{f}(x)$,

(iii) $y = \mathrm{f}(|x|)$.

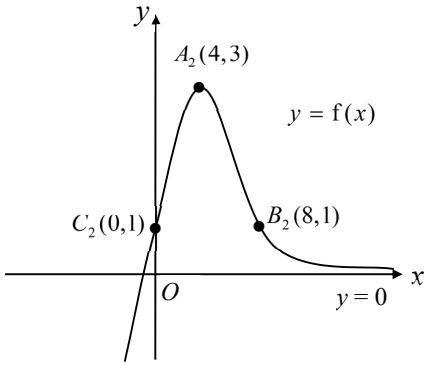
Your sketch should show clearly the coordinates corresponding to A, B and C, and the equations of asymptotes (if any).

[8]

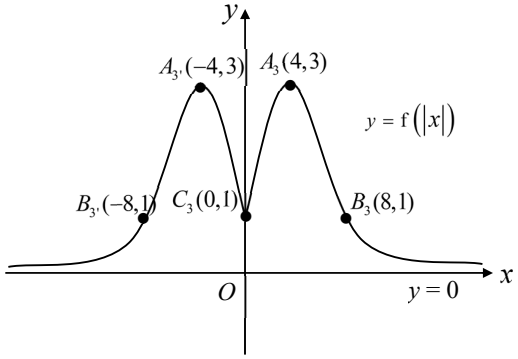
(i)



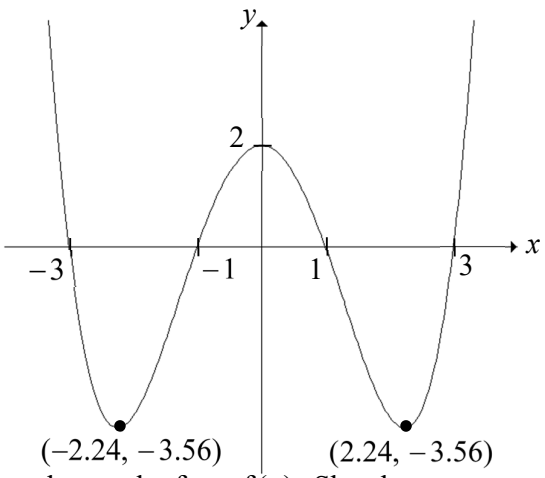
(ii)



(iii)



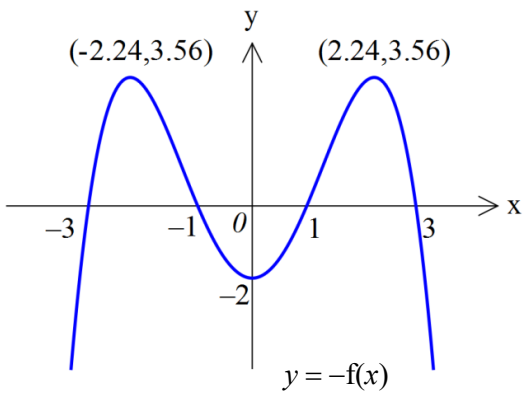
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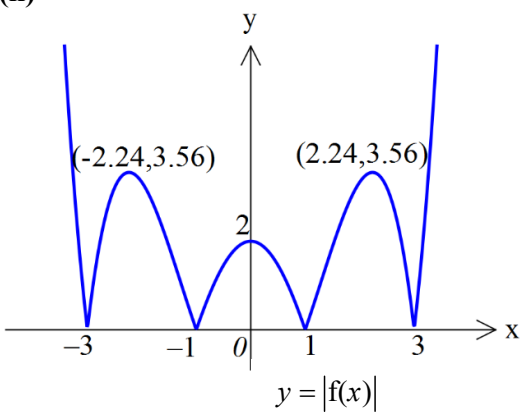
The diagram shows the graph of $y = \mathrm{f}(x)$. Sketch, on separate diagrams, the graphs of

(i) $y = -\mathrm{f}(x)$, **(ii)** $y = \left|\mathrm{f}(x)\right|$, **(iii)** $y = \mathrm{f}\left(|x| - 1\right)$, **(iv)** $y = \mathrm{f}\left(1 - |2x|\right)$. **[12]**

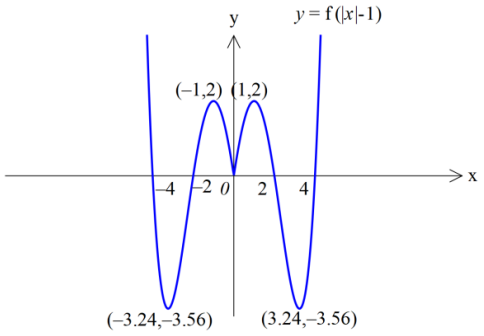
(i)



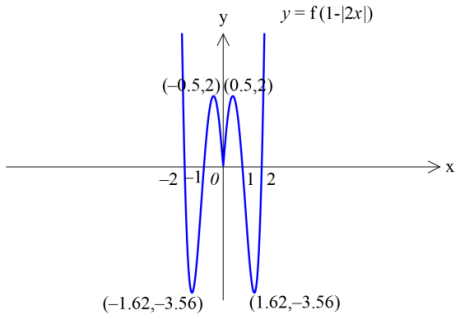
(ii)



(iii)

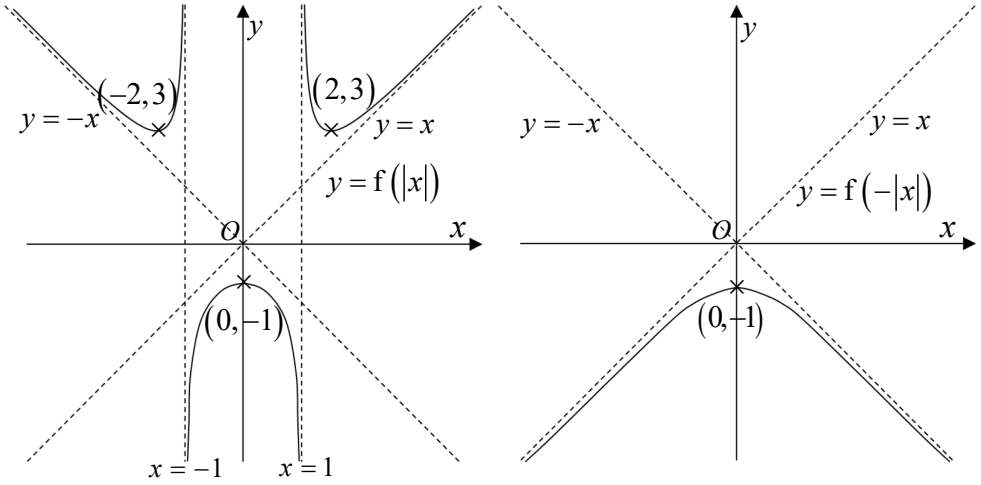


(iv)



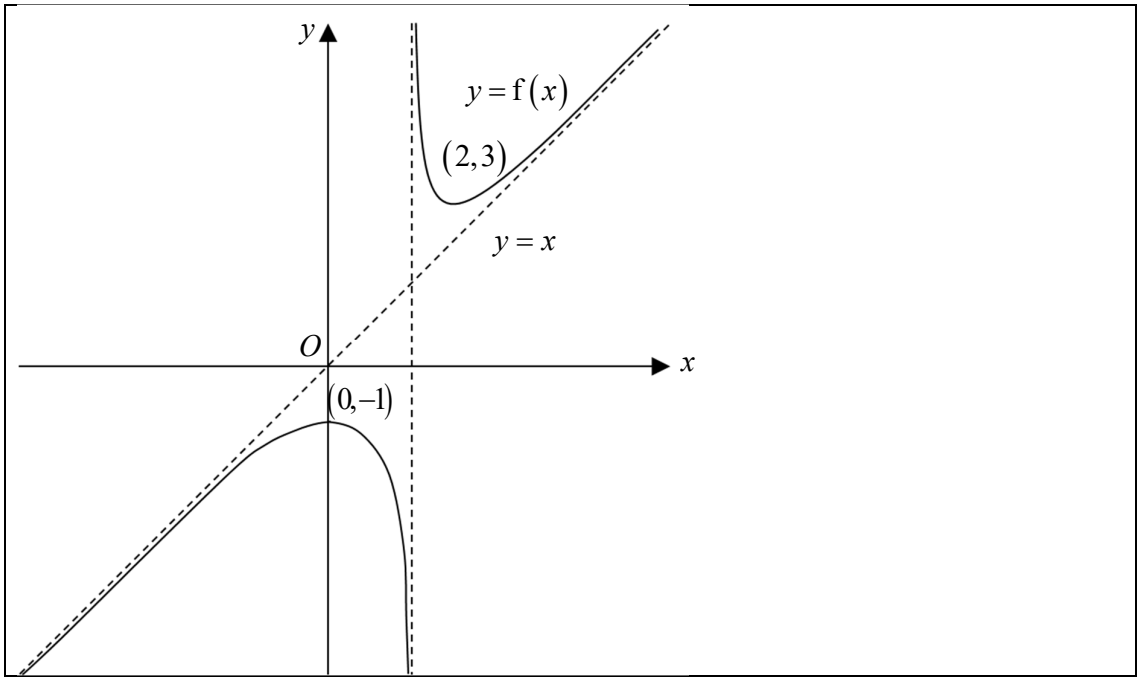
7 MI Promo 9758/2018/PU2/02/Q4a

The graphs of $y = \mathrm{f}(|x|)$ and $y = \mathrm{f}(-|x|)$ are shown below.



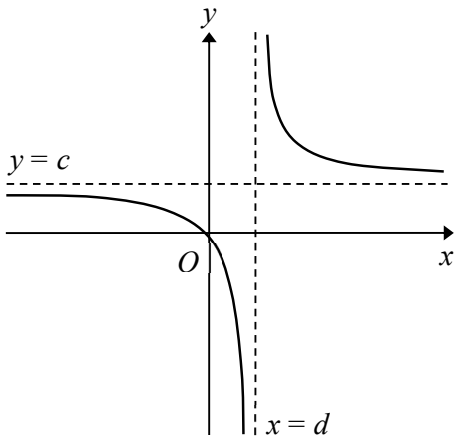
Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of any turning points and intersections with the axes. [2]

SOLUTION



8 VJC Mid-yr 9740/2006/Q11 Modified

The following shows the graph of $y = \dfrac{ax}{bx-1}$.



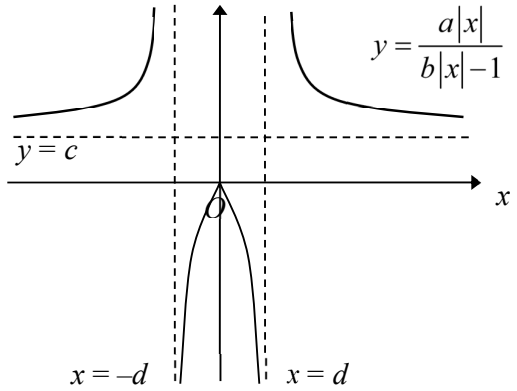
Sketch, on separate diagrams, the graphs of

(a) $y = \dfrac{a|x|}{b|x|-1}$, **(b)** $y = \left|\dfrac{ax}{bx-1}\right|$, **(c)** $y = \left|\dfrac{ax}{bx+1}\right|$.

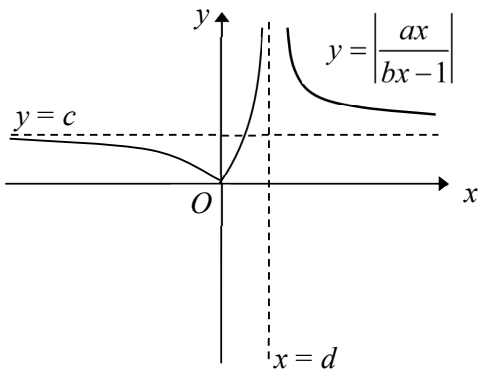
Given that $c = d = \dfrac{1}{2}$, write down the values of a and b. **[8]**

Solution

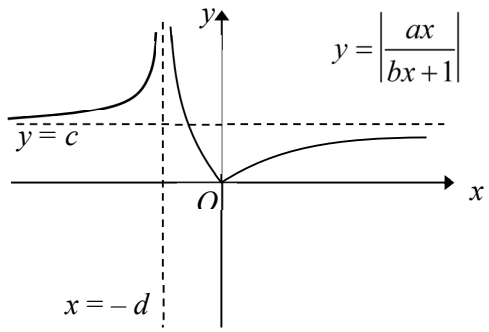
(a)



(b)



(c)



Vertical asymptote is $x = \frac{1}{2}$. Hence, when $x = \frac{1}{2}$, $\frac{1}{2}b - 1 = 0 \Rightarrow b = 2$

$$\frac{ax}{bx-1} = \frac{1}{2} + \frac{e}{2x-1} = \frac{\frac{1}{2}(2x-1)+2e}{2x-1}$$

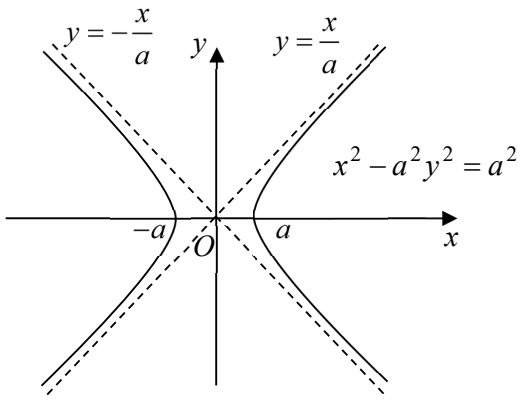
Horizontal asymptote is $y = \frac{1}{2}$. Hence, $a = 1$

Transformations involving Conic Sections

9 VJC Prelim 9740/2009/ 01/Q6

A curve C_1 has equation $x^2 - a^2y^2 = a^2$ where $a > 1$. Sketch C_1, indicating the axial intercepts, asymptotes and stationary points, if any. **[2]**

C_1 undergoes a single transformation to become C_2. Given that C_2 has a line of symmetry $y = 4$ and the point (4, 3) lies on C_2, find the value of a. **[3]**



$C_1 : x^2 - a^2y^2 = a^2$

$\frac{x^2}{a^2} - y^2 = 1$

C_2 is obtained by translating C_1 by 4 units in the positive y-direction.

$\therefore C_2$ has equation $x^2 - a^2(y-4)^2 = a^2$

When $x = 4,\ y = 3,$

$16 - a^2 = a^2$

$a^2 = 8$

$a = 2\sqrt{2}\ (\because a > 1)$

10 MI Promo 9758/2018/PU1/Q8

The curve C_1 has equation $9x^2+18x+4y^2-8y=23$.

(i) By completing the square, show that the equation of C_1 can be expressed as $\frac{(x+1)^2}{2^2}+\frac{(y-1)^2}{3^2}=1$. [2]

(ii) Sketch C_1, stating the coordinates of any points of intersection with the axes. [3]

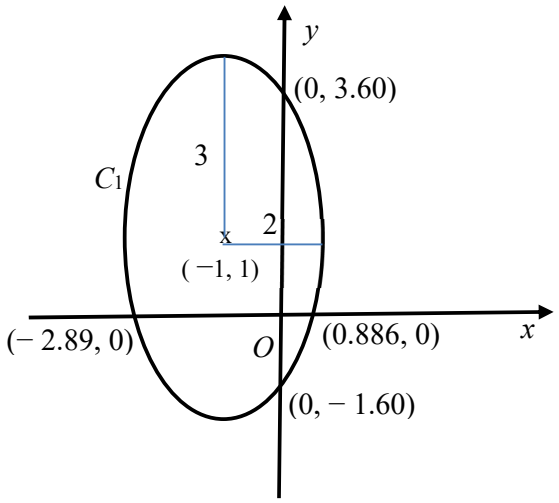
(iii) By adding a suitable graph to part **(ii)**, solve the equation $\frac{(x+1)^2}{2^2}+\frac{x^2}{3^2}=1$. [2]

(iv) State a sequence of transformations which transform the graph of C_1 to the graph of C_2 with equation $x^2+y^2=1$. [3]

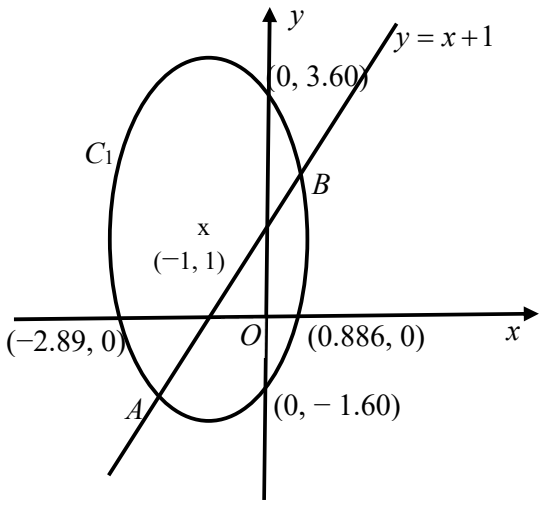
(i) [2]

$$9x^2+18x+4y^2-8y=23$$
$$\Rightarrow 9\left(x^2+2x\right)+4\left(y^2-2y\right)=23$$
$$\Rightarrow 9\left[(x+1)^2-1\right]+4\left[(y-1)^2-1\right]=23$$
$$\Rightarrow 9(x+1)^2-9+4(y-1)^2-4=23$$
$$\Rightarrow \frac{(x+1)^2}{2^2}+\frac{(y-1)^2}{3^2}=1$$

(ii) [3]



(iii) [2]



$\frac{(x+1)^2}{2^2}+\frac{x^2}{3^2}=1$. Additional graph is $y=x+1$.

By GC, x-coordinates of the intersection are –2.29 and 0.907 (3s.f.).

(iv) [3]

Method 1:

$$\frac{(x+1)^2}{2^2}+\frac{(y-1)^2}{3^2}=1$$

$$\xrightarrow{\text{(1): Replace } x \text{ by } x-1}\left(\frac{x}{2}\right)^2+\frac{(y-1)^2}{3^2}=1$$

$$\xrightarrow{\text{(2): Replace } x \text{ by } 2x} x^2+\frac{(y-1)^2}{3^2}=1$$

$$\xrightarrow{\text{(3): Replace y by y+1}} x^2+\left(\frac{y}{3}\right)^2=1$$

$$\xrightarrow{\text{(4): Replace y by 3y}} x^2+y^2=1$$

(1): Translation in the positive x-direction by 1 unit.

(2): Scaling parallel to the x-axis by scale factor $\frac{1}{2}$.

(3): Translation in the negative y-direction by 1 unit.

(4): Scaling parallel to the y-axis by scale factor $\frac{1}{3}$.

(iv)

Method 2:

$$\frac{(x+1)^2}{2^2}+\frac{(y-1)^2}{3^2}=1$$

$$\xrightarrow{\text{(1): Replace } x \text{ by } x-1} \frac{(x)^2}{2^2}+\left(\frac{y}{3}-\frac{1}{3}\right)^2=1$$

$$\xrightarrow{\text{(2): Replace } x \text{ by } 2x} x^2+\left(\frac{y}{3}-\frac{1}{3}\right)^2=1$$

$$\xrightarrow{\text{(3): Replace y by } 3y} x^2+\left(y-\frac{1}{3}\right)^2=1$$

$$\xrightarrow{\text{(4): Replace y by } y+\frac{1}{3}} x^2+y^2=1$$

(1): Translation in the positive x-direction by 1 unit.

(2): Scaling parallel to the x-axis by scale factor $\frac{1}{2}$.

(3): Scaling parallel to the y-axis by scale factor $\frac{1}{3}$.

(4): Translation in the negative y-direction by $\frac{1}{3}$ unit.

(iv)

Method 3:

$$\frac{(x+1)^2}{2^2}+\frac{(y-1)^2}{3^2}=1$$

$$\xrightarrow{\text{(1): Replace } y \text{ by } y+1} \left(\frac{x}{2}+\frac{1}{2}\right)^2+\left(\frac{y}{3}\right)^2=1$$

$$\xrightarrow{\text{(2): Replace } y \text{ by } 3y} \left(\frac{x}{2}+\frac{1}{2}\right)^2+y^2=1$$

$$\xrightarrow{\text{(3): Replace } x \text{ by } 2x} \left(x+\frac{1}{2}\right)^2+y^2=1$$

$$\xrightarrow{\text{(4): Replace } x \text{ by } x-\frac{1}{2}} x^2+y^2=1$$

(1): Translation in the negative y-direction by 1 unit.

(2): Scaling parallel to the y-axis by scale factor $\frac{1}{3}$.

(3): Scaling parallel to the x-axis by scale factor $\frac{1}{2}$.

(4): Translation in the positive x-direction by $\frac{1}{2}$ unit.

11 ACJC Prelim 9740/2011/02/Q5

The curves C_1 and C_2 have equations $9y^2 = (x+k)^2 - 9$ and $\frac{x^2}{k^2} + y^2 = 1$ respectively, where k is a real constant such that $k > 3$.

(a) Describe a sequence of transformations which transforms the graph of $x^2 - y^2 = 1$ to the graph of C_1. **[2]**

(b) (i) On the same diagram, sketch the graphs of C_1 and C_2, stating clearly the coordinates of any points of intersection with the axes and the equations of any asymptotes. **[5]**

(ii) Find the **range** of values of the positive constant a such that the equation $\frac{x^2}{a^2} + \frac{(x+k)^2 - 9}{9} = 1$ has four real roots. **[2]**

(a) C_1: $9y^2 = (x+k)^2 - 9$

$\Rightarrow \frac{(x+k)^2}{3^2} - y^2 = 1$

Sequence of transformations:

Scale the graph of $x^2 - y^2 = 1$ parallel to the x-axis by factor 3, followed by translation of k units in the negative x- direction.

OR

Translate the graph of $x^2 - y^2 = 1$ $\frac{k}{3}$ units in the negative x-direction, followed by scaling parallel to the x-axis by factor 3.

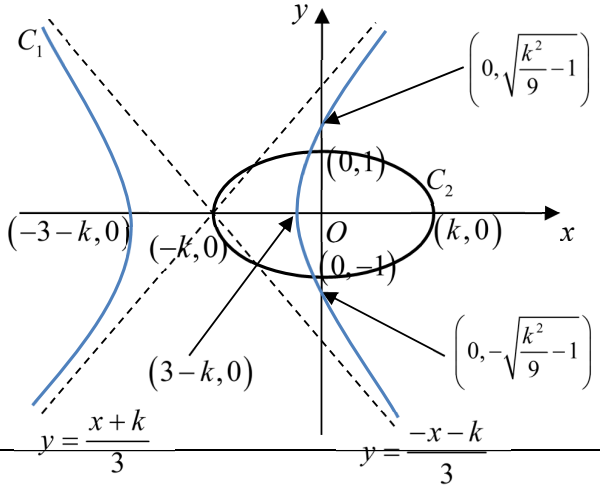
(b)(i) C_1: $\frac{(x+k)^2}{3^2} - y^2 = 1$

Intercepts: $(3-k, 0)$, $(-3-k, 0)$, $\left(0, \pm\sqrt{\frac{k^2}{9} - 1}\right)$

C_2: $\frac{x^2}{k^2} + y^2 = 1$

Intercepts: $(\pm k, 0)$, $(0, \pm 1)$

Asymptotes: $y = \pm\frac{(x+k)}{3}$ i.e. $y = \frac{x+k}{3}$ and $y = \frac{-x-k}{3}$



(ii) $\dfrac{x^2}{a^2}+\dfrac{(x+k)^2-9}{9}=1$

$\dfrac{(x+k)^2}{9}-1=1-\dfrac{x^2}{a^2}$ ------ (*)

The number of roots for equation (*) is equivalent to the number of intersection points between C_1 for general k and C_2 when $k=a$.

$\therefore$ For equation (*) to have 4 real roots, the two graphs intersect at 4 points.

Since a is a positive constant, $-a<-3-k \Rightarrow a>k+3$

12 HCI Prelim 9740/2010/01/Q12 modified

The curves C_1 and C_2 have equations $(x-2)^2 = a^2(1-y^2)$ and $y = \dfrac{x^2-4}{x+1}$ respectively, where $1 < a < 2$.

State a sequence of transformations which transforms the graph of $x^2 + y^2 = 1$ to the graph of C_1 and describe the geometrical shape of C_1. **[4]**

(i) Sketch C_1 and C_2 on the same diagram, stating the coordinates of any points of intersection with the axes and the equations of any asymptotes. **[6]**

(ii) Show algebraically that the x-coordinates of the points of intersection of C_1 and C_2 satisfy the equation $(x+1)^2(x-2)^2 = a^2(x+1)^2 - a^2(x^2-4)^2$. **[2]**

(iii) Deduce the number of real roots of the equation in part **(ii)**. **[1]**

$C_1 : \dfrac{(x-2)^2}{a^2} + y^2 = 1$

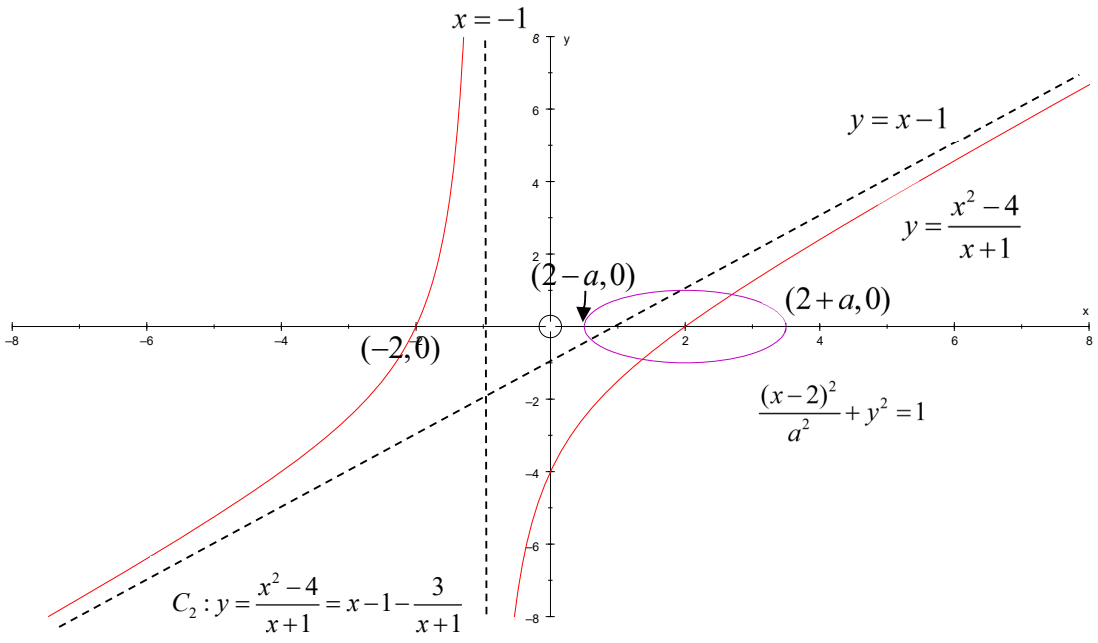
C_1 is an ellipse with centre (2,0) with the length of the semi-major axis being a units and semi-minor axis being 1 unit.

To obtain C_1 from $x^2 + y^2 = 1$, scale parallel to x-axis by a factor of a, then translate by 2 units in the positive x-direction.

OR

Translate by $\dfrac{2}{a}$ units in the positive x-direction, then scale parallel to x-axis by a factor of a

(i)

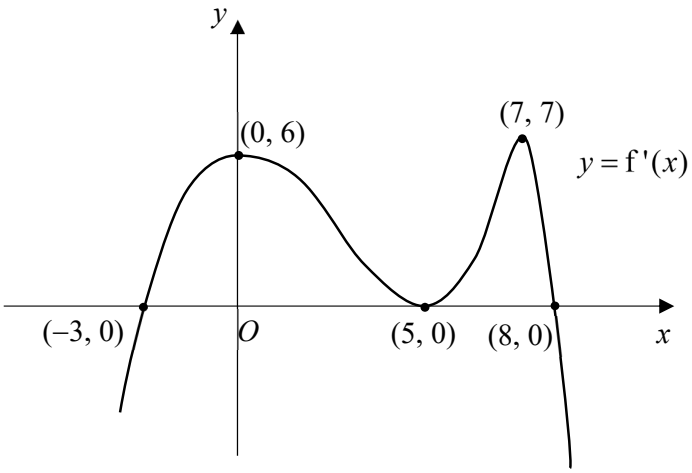


(ii) At the intersection, $(x-2)^2 = a^2\left[1-\left(\frac{x^2-4}{x+1}\right)^2\right]$ $\Rightarrow (x-2)^2 = a^2\left[\frac{(x+1)^2-(x^2-4)^2}{(x+1)^2}\right]$ $\Rightarrow (x-2)^2(x+1)^2 = a^2(x+1)^2 - a^2(x^2-4)^2$ (shown)	**(iii)** 2 real roots

Graphs of Derivative Functions & Reciprocal Functions

13 DHS Prelim 9758/2018/02/Q4 modified

The diagram below shows the curve $y = \mathrm{f}'(x)$. It has turning points at (0, 6), (5, 0) and (7, 7) and intersects the x-axis at (–3, 0) and (8, 0).



(i) State the x-coordinates of the stationary points for the curve $y = \mathrm{f}(x)$. Hence determine the nature of these stationary points. [3]

(ii) State the range of values of x such that the curve $y = \mathrm{f}(x)$ is concave downwards. [2]

(iii) A student makes a claim on the following statement:

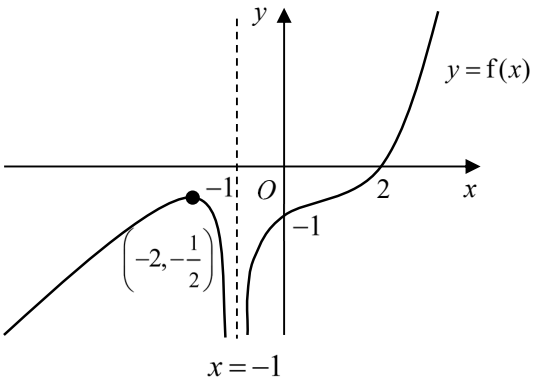
If the curve $y = \mathrm{f}(x)$ passes through the point (–3, 7), then $\mathrm{f}(x) > 7$ for $-3 < x < 5$.

Explain briefly whether the student is right to make such a claim. [1]

(i)	There are 3 stationary points at $x = -3$, 5 and 8. It has a minimum point at $x = -3$, point of inflexion at $x = 5$ and maximum point at $x = 8$.
(ii)	Range of x are either $0 < x < 5$ or $x > 7$.
(iii)	Student is right because for $-3 < x < 5$, $\mathrm{f}'(x) > 0$. This implies f is increasing for $-3 < x < 5$, thus $\mathrm{f}(x) > \mathrm{f}(-3)$, ie $\mathrm{f}(x) > 7$.

14 DHS Prelim 9740/2009/02/Q1 modified

The diagram below shows the graph of $y = \mathrm{f}(x)$. The curve has a maximum point at $\left(-2, -\frac{1}{2}\right)$ and an asymptote $x = -1$.



Sketch on separate diagrams, the graphs of

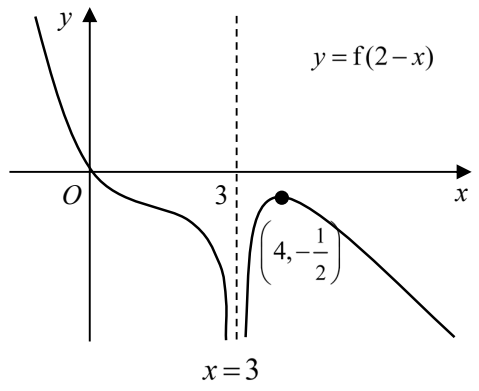
(i) $y = \mathrm{f}(2-x)$, **[2]**

(ii) $y = \dfrac{1}{\mathrm{f}(x)}$, **[3]**

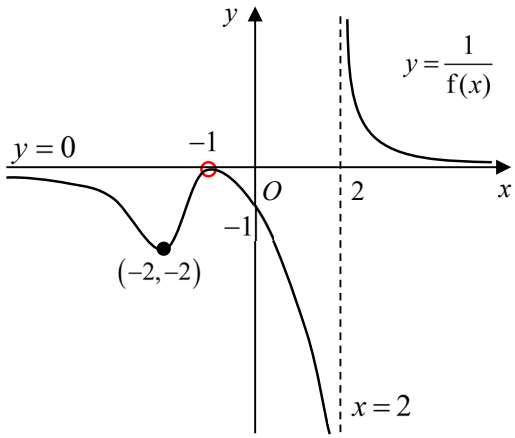
(iii) $y = \mathrm{f}'(x)$, **[3]**

showing, in each case, the coordinates of the axial intercepts, stationary points and asymptotes if they can be determined.

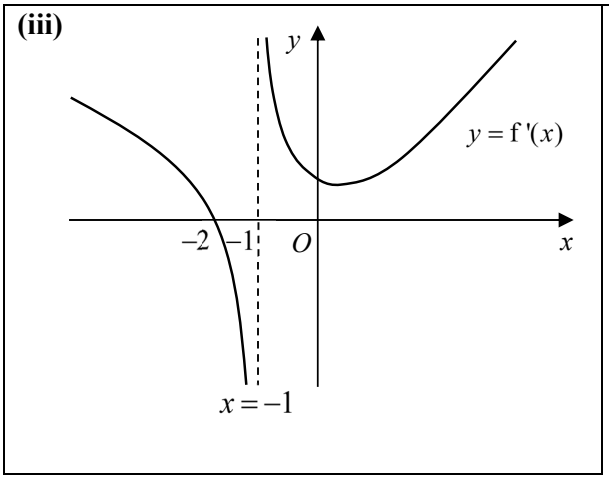
(i)



(ii)

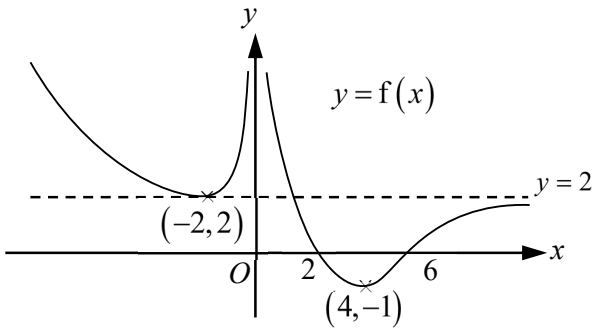


(iii)



15 DHS Promo 9740/2015/01/Q4

The diagram below shows the graph of $y = \mathrm{f}(x)$ with asymptotes $y = 2$ and $x = 0$. The curve has turning points at $(-2, 2)$ and $(4, -1)$.



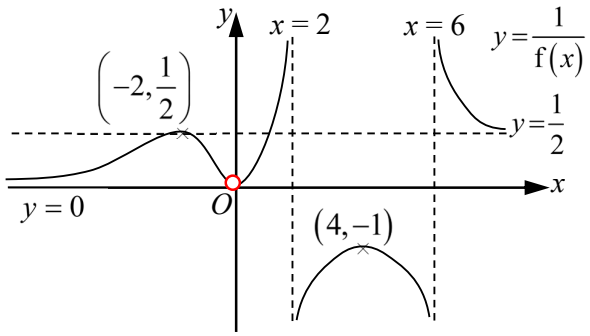
On separate diagrams, sketch the graph of

(i) $y = \dfrac{1}{\mathrm{f}(x)}$ and [3]

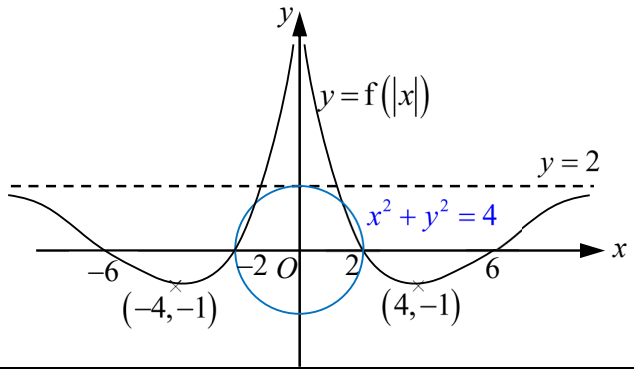
(ii) $y = \mathrm{f}(|x|)$ [2]

By adding a suitable graph to the graph in part **(ii)**, find the number of distinct roots of the equation $x^2 + \left[\mathrm{f}(|x|)\right]^2 = 4$. [2]

(i)



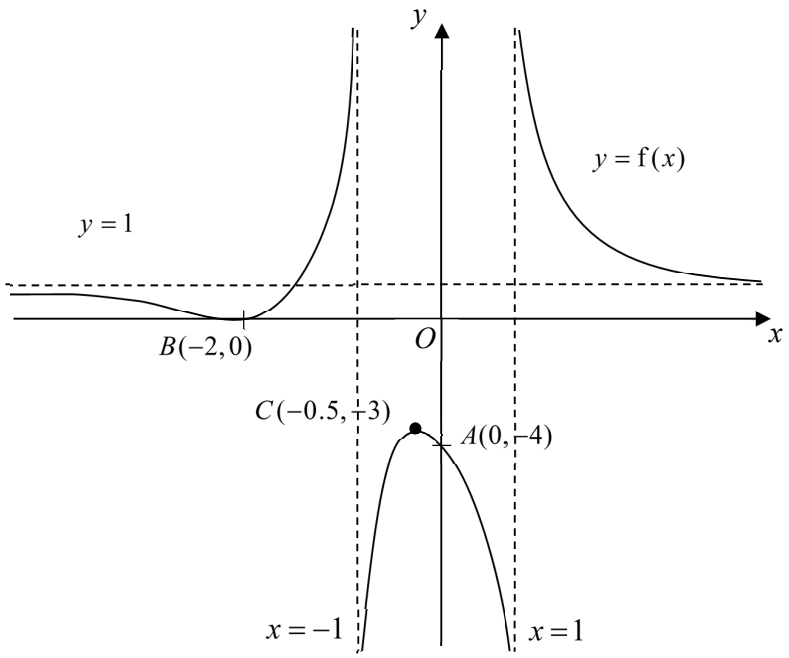
(ii)



	Consider $y=\mathrm{f}\left(\lvert x\rvert\right)$. Number of roots of $x^2+\left[\mathrm{f}\left(\lvert x\rvert\right)\right]^2=4$ is the same as the number of intersections between $x^2+y^2=4$ and $y=\mathrm{f}\left(\lvert x\rvert\right)$. From the graphs above, there are 4 intersections. Thus $x^2+\left[\mathrm{f}\left(\lvert x\rvert\right)\right]^2=4$ has 4 roots.

16 DHS Prelim 9758/2020/01/Q5

The diagram below shows the graph of $y=\mathrm{f}(x)$. It cuts the y–axis at $A\left(0,-4\right)$, has a minimum point at $B\left(-2,0\right)$ and a maximum point at $C\left(-0.5,-3\right)$. The equations of the horizontal asymptote is $y=1$ and vertical asymptotes are $x=-1$ and $x=1$ respectively. Moreover, when $x=0$, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-3$.



Sketch, on separate diagrams, the graphs of

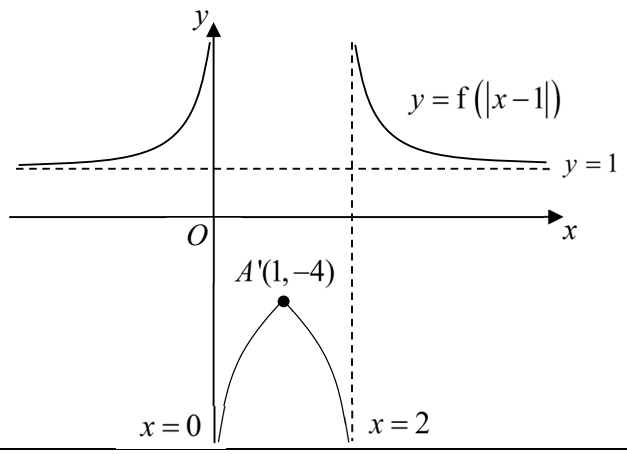
(i) $y=\mathrm{f}\left(\lvert x-1\rvert\right)$, [3]

(ii) $y=\dfrac{1}{\mathrm{f}(x)}$, [3]

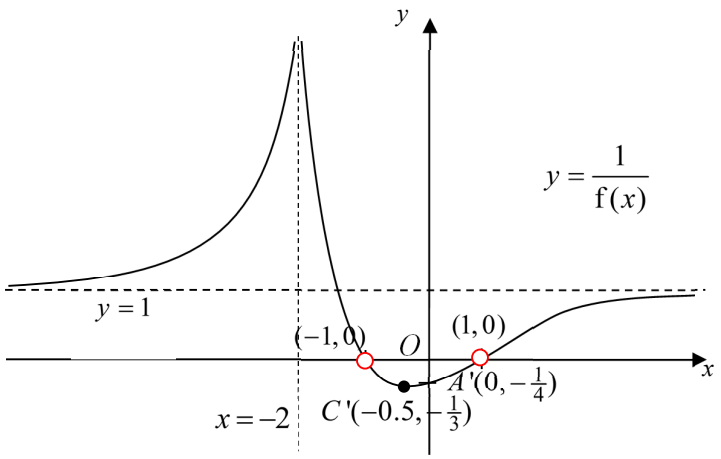
(iii) $y=\mathrm{f}'(x)$, [3]

stating clearly the equations of all the asymptotes, coordinates of any points of intersection with both axes and the coordinates of the points corresponding to A, B and C (if any and when possible).

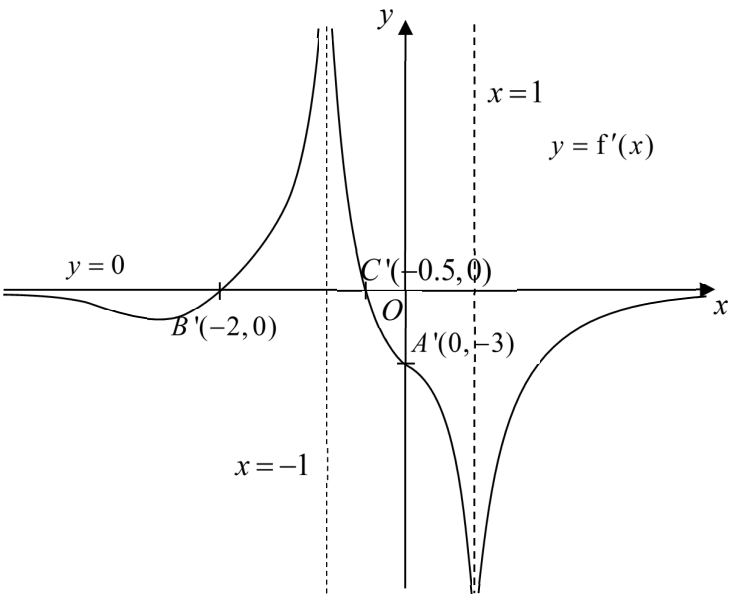
(i)


y
y = f(|x − 1|)
y = 1
O
x
A'(1, −4)
x = 0
x = 2

(ii)


y
y = 1/f(x)
y = 1
(−1, 0)
O
(1, 0)
x
A'(0, −1/4)
x = −2
C'(−0.5, −1/3)

(iii)


y
x = 1
y = f′(x)
y = 0
C'(−0.5, 0)
O
x
B'(−2, 0)
A'(0, −3)
x = −1

Questions involving Functions

17 9758/2017/02/Q3

The curve $y = \mathrm{f}(x)$ cuts the axes at $(a, 0)$ and $(0, b)$. It is given that $\mathrm{f}^{-1}(x)$ exists. State, if it is possible to do so, the coordinates of the points where the following curves cut the axes.

(i) $y = \mathrm{f}(2x)$

(ii) $y = \mathrm{f}(x-1)$

(iii) $y = \mathrm{f}(2x-1)$

(iv) $y = \mathrm{f}^{-1}(x)$ **[4]**

(i) $\left(\frac{a}{2}, 0\right)$ and $(0, b)$

(ii) $(a+1, 0)$ Note that $(1, b)$ is not on y-axis

(iii) $\left(\frac{a+1}{2}, 0\right)$ Note that $\left(\frac{1}{2}, b\right)$ is not on y-axis

(iv) $(0, a)$ and $(b, 0)$

18 9740/2012/01/Q7

A function f is said to be self-inverse if $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all x in the domain of f.

The function g is defined by $\mathrm{g}: x \to \dfrac{x+k}{x-1}$, $x \in \mathbb{R}, x \neq 1$, where k is a constant, $k \neq -1$.

(i) Show that g is self-inverse. **[2]**

(ii) Given that $k > 0$, sketch the curve $y = \mathrm{g}(x)$, stating the equations of any asymptotes and the coordinates of any points where the curve crosses the x- and y-axes. **[3]**

(iii) State the equation of one line of symmetry of the curve in part **(ii)**, and describe fully a sequence of transformations which would transform the curve $y = \dfrac{1}{x}$ onto this curve. **[4]**

(i)

If $y = \mathrm{g}(x) = \dfrac{x+k}{x-1}$, then

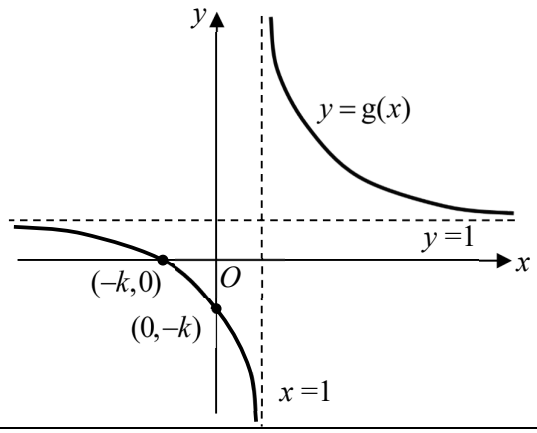
$$y(x-1) = x + k$$
$$x(y-1) = y + k$$
$$x = \frac{y+k}{y-1}$$

Hence $\mathrm{g}^{-1}(x) = \dfrac{x+k}{x-1} = \mathrm{g}(x)$ for all $x \in \mathbb{R}, x \neq 1$.

Therefore g is self-inverse.

(ii)

Let $y = \mathrm{g}(x) = \dfrac{x+k}{x-1}$ $y = \dfrac{(x-1)+k+1}{x-1} = 1 + \dfrac{k+1}{x-1}$



(iii)

Since g is self-inverse, an obvious line of symmetry is $y = x$. However, it can be seen from the sketch that $y = -x + 2$ is also a line of symmetry.

Transformations needed are:

- Scale the graph of $y = \dfrac{1}{x}$ parallel to the y-axis by a scale factor of $(k+1)$ to obtain the graph of $y = \dfrac{k+1}{x}$.

	• Translate the graph of $y = \dfrac{k+1}{x}$ in the positive x-direction by 1 unit to obtain the graph of $y = \dfrac{k+1}{x-1}$. • Translate the graph of $y = \dfrac{k+1}{x-1}$ in the positive y-direction by 1 unit to obtain the graph of $y = 1 + \dfrac{k+1}{x-1} = \dfrac{x+k}{x-1}$.